

Ex.No. 2: Implementing TDMA Slot Allocation in GSM

Objective1:

Program:

```
clc; clear; close all;
```

```
FrameTime = 4.615; Slots = 8;
```

```
SlotDuration = FrameTime/Slots;
```

```
t = 0:SlotDuration:FrameTime;
```

```
figure
```

```
subplot(2,1,1)
```

```
stem(t,ones(size(t)),'filled')
```

```
title('TDMA Time Slot Structure')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Level')
```

```
grid on
```

```
subplot(2,1,2)
```

```
plot(t,ones(size(t)),'-o','LineWidth',1.5)
```

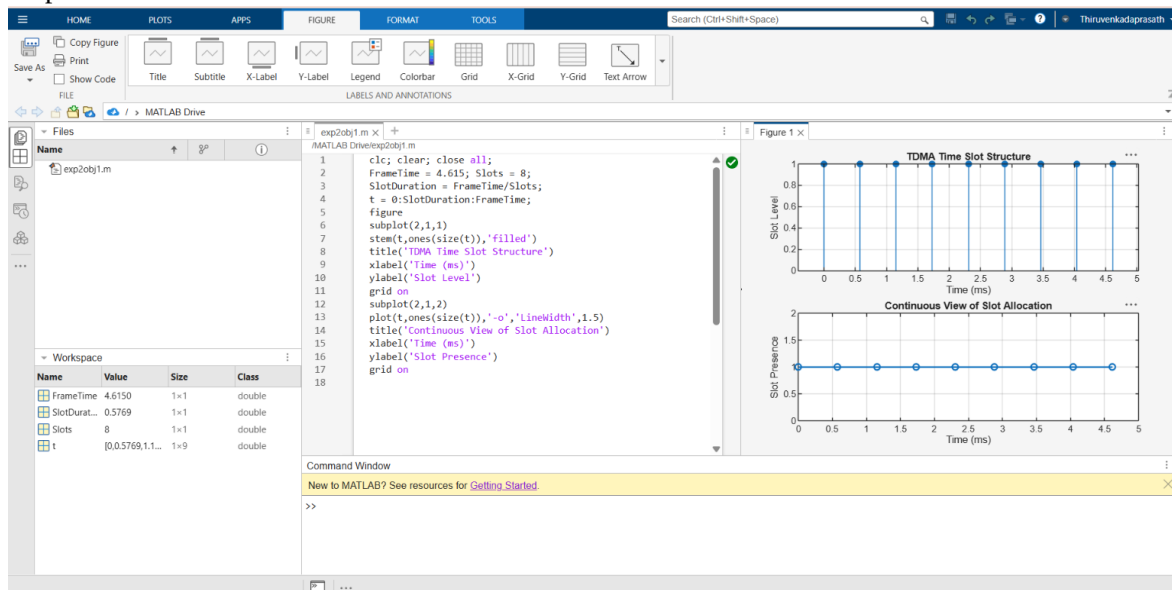
```
title('Continuous View of Slot Allocation')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Presence')
```

```
grid on
```

output:



Objective2:

Program:

```
clc; clear; close all;
```

```
TotalSlots = 8;
```

```
AllocatedSlots = 6;
```

```
FreeSlots = TotalSlots - AllocatedSlots;
```

```
Utilization = (AllocatedSlots/TotalSlots)*100;
```

```
disp('Frame Utilization (%)')
```

```
disp(Utilization)
```

```
figure
```

```
% Graph 1: Simple Line Plot (SAFE - no bar function)
```

```
subplot(2,1,1)
```

```
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
```

```
title('GSM Frame Utilization (Allocated vs Free)')
```

```
xlabel('Slot Type Index (1=Allocated, 2=Free)')
```

```
ylabel('Number of Slots')
```

```
grid on
```

```
% Graph 2: Pie Chart (SAFE)
```

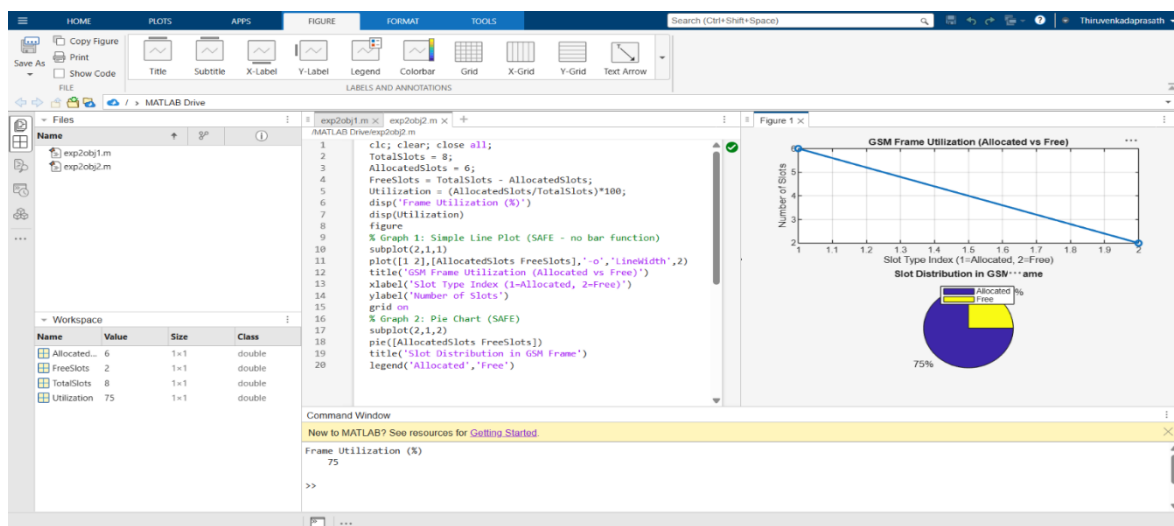
```
subplot(2,1,2)
```

```
pie([AllocatedSlots FreeSlots])
```

```
title('Slot Distribution in GSM Frame')
```

```
legend('Allocated','Free')
```

Output:



Objective3:

Program:

```
clc; clear; close all;
```

```
Users = 1:8; % Total users
```

```
TotalSlots = 5; % Available slots
```

```
Alloc = zeros(1,8); % Initialize allocation
```

```
Alloc(1:TotalSlots) = 1; % Assign slots to first users
```

```
Blocked = 1 - Alloc; % Remaining users blocked
```

```
disp('User Wise Slot Allocation')
```

```
disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Blocked)'}))
```

figure

```
% Graph 1: Allocation per user
```

```
...% Graph 2: System view (safe alternative to bar)
```

```
subplot(2,1,2)
```

```
plot(Users,Alloc,'-o','LineWidth',2)
```

```
hold on
```

```
plot(Users,Blocked,'-s','LineWidth',2)
```

```
title('TDMA Slot Utilization Analysis')
```

```
xlabel('User Index')
```

```
ylabel('Status')
```

```
legend('Allocated','Blocked')
```

```
grid on
```

Output:

